

Geetika Vadali

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RESEARCH INTERESTS

Deep Learning, Reinforcement Learning, Computer Vision, Computational Biology, Computational Neuroscience, AI for Healthcare, AI for Science, Game Theory

EDUCATION

- **Georgia Institute of Technology**

Master of Science in Computer Science

Aug. 2025 – Ongoing

Computer Vision (A), Deep Learning (A), Machine Learning (A), Autonomous Robotics (A)

Grade: **4.00/4.00**

- **Indira Gandhi Delhi Technical University for Women**

Delhi, India

BTech in Computer Science and Engineering with Artificial Intelligence

Aug. 2021 – May 2025

Grade: **9.25/10.00**

RESEARCH AND INDUSTRIAL EXPERIENCE

- **Graduate Researcher:** Georgia Institute of Technology, Metz, France

August 2025 - Currently

- Masters Thesis under Prof. Cedric Pradalier and Prof. Alex Locquet on using computer vision and deep learning techniques in reading contents of historic documents evaluated with TeraHertz signals.

- **Researcher and Participant:** Neuromatch - Computational Neuroscience Summer School 2025

July 2025

- Utilized computational modelling techniques and dynamical systems to simulate working memory as seen in human brains. Modelled the changes in the attractor-based working memory model with respect to short-term synaptic plasticity - facilitation and depression. Was primarily responsible for the programming and experimentation as part of a multi-disciplinary team.

- **Machine Learning Intern:** Aurva, Bengaluru

April 2025 - July 2025

- Tested various SOTA LLMs on data-security domain specific question-answering and compliance report generation. Employed techniques like Retrieval Augmented Generation and Prompt Engineering to handle complex and large API responses and extract the desirable context.

- **Mitacs Globalink Research Intern:** Northern Institute for Deep Learning in Ultrasound Lab, University of Alberta, Edmonton, Canada

May 2024 - August 2024

- Trained and integrated a **Deep Reinforcement Learning (DRL)** agent to an existing classifier system that facilitates earlier point-of-care diagnosis for rotator cuff tear. Was responsible for the **reward function formulation and training**. Combined concepts of video summarization, RL and medical data analysis. Advised by Prof. Abhilash R. Hareendranathan.
- Selected among 50+ interns to present my internship work as a Three-Minute Thesis for an interdisciplinary audience at Research Symposium, University of Alberta.

PUBLICATIONS

- **Terahertz reflectivity dataset: Reading text on both sides of the page:** [Link](#)

Geetika Vadali, Hugo Cherid, Erwan Émile, Haolian Shi, Leor Jacobi, DS Citrin, Cédric Pradalier, Alexandre Locquet
Data in Brief Journal August 2026, Elsevier

- **Shoulder Rotator Cuff Tear Detection from Ultrasound Videos using Deep Reinforcement Learning:** [Link](#)

Shrimanti Ghosh*, **Geetika Vadali***, Ayush Singh, Yuyue Zhou, Banafshe Felfeliyan, Assefa Wahd, Jessica Knight, Mahesh R. Panicker, Jacob L. Jaremko, Abhilash R. Hareendranathan

International Symposium on Biomedical Imaging (ISBI) 2025, Houston TX, United States

- **Video Summarization and Fracture Detection in Pediatric Wrist Ultrasound Using Deep Reinforcement Learning:** [Link](#)
Shrimanti Ghosh, **Geetika Vadali**, Yuyue Zhou, Ayush Singh, Jessica Knight, Christopher Keen, Mahesh R. Panicker, Abhilash R. Hareendranathan, Jacob Jaremko
IEEE Engineering in Medicine and Biology (EMBC) 2025, Copenhagen, Denmark
- **Spatio-Temporal Crime Detection Using Surveillance Videos in a Federated Setting:** [Link](#)
Nikita Shrivastava*, Anushka Garg*, **Geetika Vadali***, Ritika Kumari
International Conference on Computer Vision and Robotics (CVR) 2025, NIT Goa, India
- **Convergence to Nash Equilibrium: A Comparative Study of Rock-Paper-Scissors Algorithms:** [Link](#)
Geetika Vadali*, M Deekshitha Reddy*, Ritu Rani, Poonam Bansal
IEEE International Conference on Computing, Communication, and Intelligent Systems 2023, Noida, India [Code Paper](#)
- **Predicting Optimal Startup Exit Strategies with Ensemble Learning for Venture Capitalists:** [Link](#)
M Deekshitha Reddy, **Geetika Vadali**, Garima Jaiswal, Ritu Rani, and Amita Dev
Springer International Conference on Artificial Intelligence and Speech Technology 2023, IGDTUW, India.

TEACHING EXPERIENCE

- **Teaching Assistant:** Computational Neuroscience Summer School 2026, Neuromatch
July 2026
- **Graduate Teaching Assistant:** Georgia Institute of Technology, Metz, France
January 2026 - July 2026
 - Graduate Tutor for assistance and mentoring in CS 3600: Introduction to Artificial Intelligence, CS 3251: Computer Networking I, and other undergraduate CS courses without dedicated teaching assistants.

TECHNICAL SKILLS

- **Technical Languages:** Python, Java, SQL, C++, ROS2
- **Frameworks and Tools:** Tensorflow, PyTorch, Keras, Bootstrap, Pandas, Numpy, Tableau, Matplotlib, Scikit-Learn, Sympy, CI/CD, Bash Scripting
- **Courses Taken:** Machine Learning*, Autonomous Robotics*, Computer Vision (A), Deep Learning (A)

WORKSHOPS AND CONFERENCES ATTENDED

- **Conferences:** 9th Annual Conference of Cognitive Science, IIT Delhi, Dec 2022; Presented @ International Conference on Computing, Communication, and Intelligent Systems, India, Nov 2023; Upper Bound Conference, University of Alberta, Amii, May 2024; Presented @ IC-IMPACTS Legacy Conference 2024, Dec 2024
- **Workshops:** AISS - Artificial Intelligence Summer School on Federated Learning, IIIT Delhi, July 2023; Reinforcement Learning Workshop, IISc Delhi, Jan 2025; Neuromatch Computational Neuroscience Summer School, July 2025

REFERENCES

- **Prof. Cedric Pradalier:** Professor, Department of Interactive Computing, Georgia Institute of Technology, France
cedric.pradalier@gatech.edu
- **Prof. Abhilash R. Hareendranathan:** Assistant Professor, Department of Radiology, University of Alberta, Canada
hareendr@ualberta.ca